

Jacob Nguyen

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EDUCATION

University of Florida

May 2028

Bachelor of Science in Computer Engineering

GPA: 4.0/4.0

- **Relevant Coursework:** Embedded Systems, Microprocessor Applications, Advanced Systems Programming, Digital Design, Operating Systems, Data Structures & Algorithms, Digital Logic & Computer Systems

TECHNICAL SKILLS

Languages: C, C++, Rust (no_std), Python, SystemVerilog, VHDL, Assembly (RISC-V, AVR)

Embedded: NXP i.MX RT, STM32 (Cortex-M4), ESP32, AVR, FreeRTOS, Zephyr RTOS, Linux Kernel, FPGA

Peripherals: Ethernet, CAN FD, DMA, SPI, I2C, UART, ADC/DAC, Timers/PWM

Tools: Vim, Git, Linux, Bash, GDB, Make, Yocto, Analog Discovery 3 (Oscilloscope, Logic Analyzer)

EXPERIENCE

Embedded Software Engineer Intern

August 2026 – Present

NASA Armstrong Flight Research Center

Edwards AFB, CA

- Built a custom Yocto BSP for a Microchip PolarFire RISC-V SoC with device tree, U-Boot, and partition changes
- Developed cFS flight software modules in C for NASA's spaceflight computing platform targeting RISC-V
- Engineered embedded firmware in C for NASA's Raven eVTOL research aircraft on STM32 using Zephyr RTOS

Firmware Engineer

October 2025 – May 2026

Gator Motorsports (FSAE), University of Florida

Gainesville, FL

- Wrote a register-level CAN FD driver in no_std Rust for the NXP i.MX RT1176 replacing C SDK wrappers
- Implemented async transmit using semaphore-gated mailbox pool with fair FIFO ordering across 7 TX buffers
- Built interrupt-driven receive pipeline with Global Unlock (TIMER read) to prevent hardware mailbox stalls
- Ported NXP SDK CAN FD bit timing algorithm to Rust for automatic dual-rate prescaler and segment calculation
- Designed bitfield-based register abstraction layer mapping all FlexCAN control, timing, and status registers

Teaching Assistant, Digital Logic

January 2026 – May 2026

Herbert Wertheim College of Engineering, University of Florida

Gainesville, FL

- Debugged breadboard wiring, timing violations, and FPGA logic errors for 15+ students during office hours
- Taught weekly lab section of 10 students through progressive digital design labs from basic gates to a CPU
- Wrote and graded custom weekly quizzes assessing student comprehension before hands-on hardware labs

PROJECTS

Baremetal RC Car – STM32, C

April 2026

- Implemented closed-loop PI speed control with feedforward estimation, anti-windup clamping, and 100Hz update
- Built dual-wheel encoder feedback using TIM2 input capture with edge priming and SysTick stopped-wheel watchdog
- Wrote NEC IR decoder using timer input capture and DMA to hardware-capture 66 protocol edges at 1 MHz
- Resolved DMA/CPU flag race by cross-mapping a second capture channel to the same pin as a dedicated IRQ source
- Designed 4-channel 20 kHz PWM motor control with 9-level speed mapping from decoded IR remote commands
- Wrote modular driver library (GPIO, Timer, PWM, DMA, IR, UART) with typed enums and config structs in C

VGA Pong – SystemVerilog, DE10-Lite

March 2026

- Designed 2-player real-time game in SystemVerilog with 640x480 VGA output at 60Hz using 25MHz pixel clock
- Engineered FSM-driven game states and pixel-coordinate collision detection across 7 concurrent user inputs
- Used BRAM-based ROMs for bitmap storage with pipeline register compensation for 1-cycle read latency

SwampRise Bus Alarm – ESP32, C++, ESP-IDF, FreeRTOS

January 2026

- Won “Best Hardware Integration” at SwampHacks with smart alarm syncing live bus data to RTC and displays
- Implemented dual-core FreeRTOS task management for concurrent API polling, display refresh, and web serving
- Interfaced I2C real-time clock and SPI OLED display modules to ESP32 hardware pins with custom driver logic